

Minh Quan (Andy) Quach

(226) 883-4546 | Email | LinkedIn | GitHub | Portfolio

SUMMARY

Data Engineer with a strong software engineering background and hands-on experience building ETL pipelines, cloud-based data solutions, and analytics-ready datasets. Skilled in Python, SQL, Snowflake, AWS, Airflow, and data modeling, with experience designing reliable data pipelines, implementing data quality checks, and supporting analytical use cases through scalable data architectures.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Data Engineering: ETL, Data Modeling, Dimensional Modeling, Data Quality, Data Validation, PySpark, Apache Spark, Databricks

Cloud & Big Data: AWS (S3, Glue, Athena, Lambda, EventBridge)

Orchestration: Airflow, Prefect

Databases: PostgreSQL, MySQL, Snowflake

Tools: Git, Bash, Linux, Docker (basic), REST APIs, Jira, Confluence

DATA PROJECTS

2025-Present

Multi-Source Crypto Data Pipeline (Python + Snowflake + Prefect)

2026

- Built an end-to-end pipeline to ingest cryptocurrency data from multiple APIs (CoinGecko, Kraken)
- Designed and maintained analytical data models to support cross-source cryptocurrency analysis
- Implemented incremental loading using Snowflake MERGE to avoid full table reloads and improve pipeline efficiency
- Orchestrated scheduled execution with Prefect (5-minute intervals) to support near real-time data updates
- Developed analytical SQL queries to compare price discrepancies across sources
- Added validation checks for schema consistency and missing values to improve data reliability

Multi-Source E-commerce ETL Pipeline (AWS S3 + Glue + Athena + Lambda)

2026

- Built a serverless ETL pipeline on AWS to ingest transaction (CSV) and product API data into an S3 data storage
- Implemented API ingestion using AWS Lambda to fetch and store semi-structured JSON data in S3
- Transformed raw data into partitioned Parquet datasets using AWS Glue (PySpark)
- Designed dimensional models (fact_transactions, dim_products) to support scalable analytical reporting
- Developed analytical SQL queries in Athena to generate insights on revenue, top products, and category performance
- Implemented data validation during PySpark transformations to ensure analytics-ready datasets

ETL Log Pipeline (Python + PostgreSQL + Airflow)

2025

- Built an ETL pipeline to process 1.8M+ web server log records, transforming unstructured data into queryable format for analytics
- Parsed unstructured logs into structured schema (IP, timestamp, endpoint, status)
- Implemented bulk loading with PostgreSQL COPY into staging tables
- Designed staging → upsert pipeline with deduplication and idempotent processing to ensure data reliability and prevent duplicate ingestion
- Orchestrated workflows using Airflow DAGs and cron scheduling
- Wrote unit tests and validation checks for log parsing and deduplication logic

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Elinext

2015 – 2023

- Developed large-scale payment-enabled mobile applications supporting customer transactions across multiple international markets

- Worked extensively with transactional workflows, customer account data, payment integrations, and operational reporting systems
- Integrated APIs responsible for payment processing, account management, parking sessions, and billing-related events
- Partnered with backend teams to validate data integrity, business rules, and cross-system consistency across distributed services
- Supported multi-market deployments with country-specific payment providers and localized business requirements

EDUCATION

Master of Engineering (MEng), Electrical & Computer Engineering	2024
University of Waterloo — Waterloo, ON	
Specialization: Software Average: 87%	
Bachelor of Computer Science	2015
University of Science — Ho Chi Minh City Average: 84%	

CERTIFICATIONS

IBM Data Engineering Professional Certificates — Coursera
 AWS Certified Cloud Practitioner (CLF-C02) — Credential
 Microsoft Certified: Power BI Data Analyst Associate (PL-300) — In Progress